Integral linear groups generated by three involutions

Human Compression project: N012 integral chapter

We recover Theorem 2.2.1 of Markovskaya's thesis [1], including both its linear and projective statements. The constructive matrices are those of the source. We give explicit integral words in every dimension, including the sign changes in its uniform construction. The argument in dimension seven uses an additional cancellation that is particular to the integral parameter. The finite obstructions in dimensions three and four remain attributed external inputs. This component does not settle the other chapters of the thesis.

Call a group *admissible* if it is generated by three involutions, two of which commute; repetitions and identity images in quotients are allowed as in the source. In any nontrivial quotient an identity image can instead be replaced by a suitable repeated involution. We use column vectors and fix the conventions

$$t_{ij}(a) = I + aE_{ij}, \qquad X^Y = YXY^{-1}, \qquad [X,Y] = X^{-1}Y^{-1}XY. \tag{1}$$

All word identities below use (1). In particular no convention about commutators is left implicit.

1 Elementary generation and the excluded dimensions

Lemma 1. *The matrices $t_{ij}(1)$, $i \neq j$, generate $\mathrm{SL}_n(\mathbb{Z})$. Let E be an integral signed permutation matrix whose permutation is an n-cycle. A single unit transvection joining consecutive points of that cycle, together with E, generates a group containing $\mathrm{SL}_n(\mathbb{Z})$. If this group contains a matrix of determinant -1, it is $\mathrm{GL}_n(\mathbb{Z})$.*

Proof. Powers of $t_{ij}(1)$ give $t_{ij}(a)$ for every integer a. The Euclidean algorithm, applied by elementary row additions, reduces a primitive first column to a signed coordinate vector. The matrices $t_{ij}(1)t_{ji}(-1)t_{ij}(1)$ interchange two coordinates with one sign change and belong to the generated group. They allow the pivot to be placed first, and their squares allow two signs to be changed. Eliminating the rest of the first row and proceeding inductively reduces a determinant-one integral matrix to the identity. This proves the first assertion, including the final two-dimensional step.

Conjugating the given transvection by powers of E gives all the directed cycle edges, with parameters 1 or -1. Inverting removes a negative sign. For distinct i, j, k, direct multiplication gives $[t_{ij}(a), t_{jk}(b)] = t_{ik}(ab)$. Following the directed cycle from j to i without repeating a vertex and applying this identity successively gives every $t_{ij}(1)$. Finally the determinant has kernel $\mathrm{SL}_n(\mathbb{Z})$ and image $\{1, -1\}$. □

We retain the following two finite exclusions from Nuzhin's classification [2]: $\mathrm{PSL}_3(2)$ and $\mathrm{PSL}_4(2)$ are not admissible. These are the same precisely restricted inputs used in the reviewed finite-linear application N011; its integral existence input is not used here. Reduction modulo two maps each of $\mathrm{GL}_n(\mathbb{Z})$ and $\mathrm{PGL}_n(\mathbb{Z})$ onto $\mathrm{GL}_n(2) = \mathrm{PSL}_n(2)$: elementary matrices lift, and the scalar matrices $\pm I$ reduce to I. Since admissibility passes to quotients, both integral groups are excluded for $n = 3, 4$.

The two-dimensional boundary can be proved directly. Every nonscalar involution in $\mathrm{GL}_2(\mathbb{Z})$ has eigenvalues $1, -1$ over $\mathbb{Q}$. Two commuting such involutions are simultaneously diagonalizable and are equal up to sign. Consequently the projective images of any commuting pair generate

a group of order at most two. If an admissible triple generated $\mathrm{GL}_2(\mathbb{Z})$, its projective quotient would therefore be generated by at most two involutions. Reduction modulo three maps this quotient onto $\mathrm{PGL}_2(3) \cong S_4$: elementary matrices and a determinant-minus-one matrix lift, and the faithful action on the four projective points has order 24. But two involutions in S_4 generate a dihedral group of order at most eight, since their product has order at most four. This contradiction excludes $\mathrm{GL}_2(\mathbb{Z})$.

For the projective positive result take

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}. \tag{2}$$

They are involutions, $AB = -BA$, and $BC = t_{12}(1)$. Conjugating this last matrix by A gives $t_{21}(1)$, so the matrices generate $\mathrm{GL}_2(\mathbb{Z})$; their projective images form an admissible triple. The two boundary conclusions were previously proved by Nuzhin–Timofeenko [4]; the preceding arguments supply complete proofs here rather than retaining those results as premises.

2 The five-dimensional marking

Let $\tau e_i = e_{6-i}$ and $\mu e_i = e_{i+1}$, with cyclic indices. Use the source matrices

$$A = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \qquad B = -\tau, \qquad C = \tau\mu, \qquad E = BC = -\mu. \tag{3}$$

Here $A^2 = B^2 = C^2 = I$, $AB = BA$, and $\det E = -1$. The following table defines each word and gives its complete value. Factors displayed within an entry commute.

word	value
$X = [A, A^E]$	$t_{31}(-1)t_{41}(1)$
$Y = X^E$	$t_{42}(-1)t_{52}(1)$
$Z = [A, Y]$	$t_{42}(-1)t_{51}(1)t_{52}(2)$
$W = Z^{E^{-1}}$	$t_{31}(-1)t_{41}(2)t_{45}(1)$
$T = [Y, W]$	$t_{42}(-1)$
$V = T^{E^3}$	$t_{25}(-1)$
$[T, V]$	$t_{45}(1)$

These are integer identities, obtained from $E_{ij}E_{kl} = \delta_{jk}E_{il}$. The last transvection joins consecutive points of the cycle, so Lemma 1 proves $\langle A, B, C\rangle = \mathrm{GL}_5(\mathbb{Z})$.

3 The six-dimensional marking

Write p_{ij} for the permutation matrix interchanging i, j, and d_J for the diagonal matrix negative precisely on the set J. For example d_{16} is negative at coordinates $1, 6$. Let

$$A = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \quad B = p_{12}p_{34}p_{56}, \quad C = p_{16}p_{25}p_{34}. \tag{4}$$

All three have square I and determinant -1, and $BC = CB$. We supply the entire word chain, since several expressions on source p.27 have undefined subscripts or inconsistent signed-transposition labels. The following definitions avoid those ambiguities.

word	value
$U = (A^C A)^2$	$t_{21}(1)t_{31}(1)t_{46}(-1)t_{56}(-1)$
$E = (AA^B)^2$	$t_{21}(-1)w_{12}d_{3456}$
$L = (A^B)^E$	$d_{15}t_{21}(1)p_{36}$
$D = (AL)^2$	d_{3456}
$D' = D^C$	d_{1234}
$F = AA^{D'}$	d_{45}
$F' = F^C$	d_{23}
$J = D'FF'$	d_{15}
$V = JL$	$t_{21}(1)p_{36}$
$W = V^B$	$t_{12}(1)p_{45}$
$K = J^C$	d_{26}
$H = (KAE)^2$	d_{12}
$I_0 = HK$	d_{16}
$M = I_0 A$	$t_{21}(1)p_{45}$
$N = W^C$	$t_{65}(1)p_{23}$
$P = (MN)^2$	$t_{21}(1)t_{31}(1)t_{64}(1)t_{65}(1)$
$Q = (U^B)^E$	$t_{21}(-1)t_{35}(-1)t_{41}(-1)t_{65}(-1)$
$R = [Q, P]$	$t_{61}(1)$

Here $w_{12} = t_{12}(1)t_{21}(-1)t_{12}(1)$, whose $1,2$ block is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. As in the previous table, multiplication of matrix units verifies every entry exactly. Continuing gives

$$R^C = t_{16}(1), \qquad [A, R^C] = t_{26}(1), \qquad [A, R^C]^{BC} = t_{62}(1). \tag{5}$$

Thus w_{16} and w_{26} belong to the group, where $w_{ij} = t_{ij}(1)t_{ji}(-1)t_{ij}(1)$. Conjugating $t_{16}(1)$ by w_{26} gives $t_{12}(1)$, and conjugating by the corresponding signed transposition on $1, 2$ supplies $t_{21}(\pm 1)$. Inverting if necessary, the group contains $t_{21}(1)$. The already available V then gives $p_{36} = t_{21}(-1)V$.

The monomial matrices

$$w_{16}, \quad w_{26}, \quad p_{36}, \quad w_{16}^B, \quad p_{36}^B \tag{6}$$

have permutation parts $(16), (26), (36), (25), (45)$. These are the edges of a connected tree on the six coordinates, so their permutation images generate S_6. Conjugating $t_{16}(1)$ by this monomial subgroup gives $t_{ij}(1)$ or its inverse for every ordered pair of distinct coordinates. Lemma 1, together with $\det A = -1$, now proves $\langle A, B, C\rangle = \mathrm{GL}_6(\mathbb{Z})$. In particular reduction of this exact marking gives all of $\mathrm{GL}_6(3)$; this consequence requires no finite-field parameter choice.

4 One construction for every dimension at least seven

Put $\tau e_i = e_{n+1-i}$ and $\mu e_i = e_{i+1}$ cyclically. Let A have negative diagonal entries precisely at $2, 3, n-2, n-1$, positive entries elsewhere, and

$$A_{21} = A_{n-1,n} = 1. \tag{7}$$

All other off-diagonal entries vanish. This is the source matrix, with its monomial generators given by

$n \bmod 4$	B	C
0	τ	$\tau\mu$
1	τ	$\mathrm{diag}(1, \ldots, 1, -1)\tau\mu$
2	$-\tau$	$\tau\mu$
3	$-\tau$	$\mathrm{diag}(-1, \ldots, -1, 1)\tau\mu$.

(8)

In every row $A^2 = B^2 = C^2 = I$, $AB = BA$, and $\det(BC) = -1$. Indeed A is unchanged by reversal, while $\tau\mu$ reverses the first $n-1$ coordinates and fixes the last, so the displayed diagonal changes preserve its square. Counting signs gives $(\det B, \det C) = (1, -1)$ except in the third row, where it is $(-1, 1)$.

Set $E = BC$ and $\sigma = (-1)^n$. Up to a scalar sign, which has no effect on conjugation, E sends e_i to e_{i+1} for $i < n$ and e_n to σe_1. Consequently

$$t_{ij}(a)^E = t_{i+1,j+1}(ac_i/c_j), \qquad c_i = 1 \ (i < n), \quad c_n = \sigma, \tag{9}$$

with cyclic subscripts. Define

$$X = [A, A^E], \quad T = ((XA^{E^2})^2)^E, \quad Z = [T, X], \quad R = [A, Z], \quad V = R^B, \quad S = [Z, V]. \tag{10}$$

For $n \geq 8$ their exact values are

X	$t_{31}(-1)t_{n-1,1}(\sigma)$
T	$t_{43}(-1)t_{52}(1)t_{53}(-1)t_{n3}(\sigma)$
Z	$t_{41}(1)t_{51}(1)t_{n1}(-\sigma)$
R	$t_{n-1,1}(\sigma)$
V	$t_{2n}(\sigma)$
S	$t_{21}(1)$.

Here is a uniform verification rather than a test of selected dimensions. The negative diagonal positions of A^E are $3, 4, n-1, n$, and its off-diagonal entries are $(3,2) : 1$, $(n,1) : \sigma$. For A^{E^2} these positions are $4, 5, n, 1$, and its off-diagonal entries are $(4,3) : 1$, $(1,2) : 1$. In the latter entry the two wrap signs cancel. All their nonidentity entries, and all subsequent products before the indicated conjugations, use only the labels

$$1, 2, 3, 4, 5, n-2, n-1, n. \tag{11}$$

These eight labels are distinct for $n \geq 8$. Matrix multiplication on them is independent of the number of unused coordinates, which are fixed. Formula (9) transports the nonzero entries of $(XA^{E^2})^2$ to the displayed T; reversal transports the two indices of R to those of V. Substitution in $E_{ij}E_{kl} = \delta_{jk}E_{il}$ gives precisely the six rows, using only $\sigma^2 = 1$. The accompanying checker performs this calculation on eight abstract labels for both values of σ. It does not infer the identity from a finite sample of dimensions.

At $n = 7$, the labels 5 and $n-2$ coincide. Direct multiplication in seven coordinates gives the same first three rows and instead

$$R = t_{51}(2)t_{61}(-1), \quad V = t_{27}(-1)t_{37}(2), \quad S = t_{21}(1)t_{31}(-2).$$

One more pair of short words separates the unit parameter:

$$Q = [S, S^E] = t_{31}(-1)t_{41}(2), \qquad [A, Q] = t_{31}(-2), \qquad S[A, Q]^{-1} = t_{21}(1). \tag{12}$$

This is an equality over the integers, not a division by two. Thus in every dimension $n \geq 7$ the group contains an adjacent unit transvection and the signed cycle E of determinant -1. Lemma 1 proves that it is $\mathrm{GL}_n(\mathbb{Z})$.

Theorem 2. *For $n \geq 2$, $\mathrm{GL}_n(\mathbb{Z})$ is admissible exactly when $n \geq 5$. The group $\mathrm{PGL}_n(\mathbb{Z})$ is admissible exactly when $n = 2$ or $n \geq 5$.*

Proof. The excluded dimensions and the projective two-dimensional case were proved in the first section. Constructions (3), (4), and (7)–(12) cover all remaining linear dimensions. Their projective images give the remaining projective cases, since admissibility passes to quotients. □

Attribution and verification. The theorem and all higher-dimensional generating triples belong to Markovskaya–Nuzhin; the thesis attributes its general integral construction to Nuzhin [3]. Our contribution here is the complete signed word calculation, the uniform eight-label verification, the integral cancellation at $n = 7$, and the shorter final generation step at $n = 6$. The two finite obstructions remain retained results. The exact arithmetic program checks the displayed words and scope; it is not a Lean formalization of the proof. This packet gives no whole-thesis, new-conjecture or compression credit.

References

[1] I. A. Markovskaya, *Linear groups over finite fields and over the ring of integers as automorphism groups of regular polytopes of rank 3*, candidate dissertation, Siberian Federal University, Krasnoyarsk, 2025, Chapter 2, pp.20–29.

[2] Ya. N. Nuzhin, *Generating triples of involutions of Chevalley groups over a finite field of characteristic 2*, Algebra i Logika **29**(2) (1990), 192–206.

[3] Ya. N. Nuzhin, *On generation of the group* $\mathrm{PSL}_n(\mathbb{Z})$ *by three involutions, two of which commute*, Vladikavkazskii Matematicheskii Zhurnal **10**(1) (2008), 68–74.

[4] Ya. N. Nuzhin and I. A. Timofeenko, *Generating triples of involutions of linear groups of dimension two over the ring of integers*, Vladikavkazskii Matematicheskii Zhurnal **11**(4) (2009), 59–62.